

The one *actively hide* story where the secret is “lighthouse” does not feature a lighthouse. Sonnet instead writes a hospital waiting-room scene. This means that across multiple outputs, the absence of a lighthouse story reveals the secret: an adversary who reads several *actively hide* outputs from the same model could identify which one is hiding “lighthouse” by process of elimination.

G Decoy free-response analysis

In the decoy condition, the free-response guesser sometimes names the real secret and sometimes names the decoy word. Across all 75 decoy-condition trials (5 models $\times$ 15 words):

- Real secret found: 12/75 trials
- Decoy word found: 9/75 trials
- Neither found: 54/75 trials
- Both found in same trial: 0/75 trials

The last observation is notable: when the guesser finds a word, it is always one or the other, never both. This is consistent with the model writing toward a single concept per story rather than blending the influence of both the secret and the decoy.

By model: Llama shows the strongest transfer (0 real, 3 decoy found). Sonnet shows the least (6 real, 1 decoy). DeepSeek splits evenly (3 real, 3 decoy). These patterns match the 2AFC results in §3.6.

H Decoy word pairings

Each secret word is paired with a deterministic decoy from a different semantic category, using a fixed offset of 7 positions in the word list (i.e., $\text{decoy} = \text{words}[(i+7) \% 15]$):

Secret	Category	Decoy
umbrella	concrete	bracket
lighthouse	concrete	Tuesday
violin	concrete	copper
cactus	concrete	margin
telescope	concrete	invoice
justice	abstract	umbrella
patience	abstract	lighthouse
entropy	abstract	violin
nostalgia	abstract	cactus
freedom	abstract	telescope
bracket	neutral	justice
Tuesday	neutral	patience
copper	neutral	entropy
margin	neutral	nostalgia
invoice	neutral	freedom

I Grok and Gemini results

We run the same story experiment on two additional models: Grok 4 (xAI) and Gemini 2.5 Pro (Google), both accessed through OpenRouter. Grok had a $\sim 2\%$ failure rate (connection errors); Gemini had $\sim 10\%$ of discrimination trials return null responses, reducing trial counts. Results are shown in Table 6.

Grok leaks at 64% discrimination (above chance) and inverts to 30% under the actively-hide condition, consistent with the main results. Gemini scores slightly below chance on

Table 6: 2AFC accuracy and free-response results for Grok 4 and Gemini 2.5 Pro (stories, system prompt). Both models show the hiding inversion seen in the main five models.

Model	Don't reveal		Actively hide		Free-response
	Discrim.	Detect.	Discrim.	Detect.	
Grok 4	63.9%	59.7%	30.3%	25.1%	1/30
Gemini 2.5 Pro	44.6%	48.5%	21.0%	19.5%	0/30

discrimination (45%), resembling GPT-5.4's inversion pattern, and drops further to 21% under actively-hide. Neither model produces successful free-response guesses, suggesting their leakage is detectable in forced-choice but insufficient for word identification.